ELECTRIC VEHICLE MANAGEMENT SYSTEM USING ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

INDIVIDUAL THESIS

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ABSTRACT

Electric vehicle (EV) drivers face range uncertainty, unpredictable queues at charging stations and limited visibility of charging cost, while existing applications typically address routing, station discovery or range estimation separately. This thesis reports the individual component of the SmartEV project, Intelligent EV Charging Optimization and Range Prediction, which integrates range prediction, queue forecasting, station recommendation, route feasibility, charging-time estimation and cost analytics in one decision-support system. A Design Science Research approach with experimental evaluation was followed. Four range models (Linear, Ridge, Random Forest, Gradient Boosting) and three multi-output queue models (Ridge, Random Forest, Gradient Boosting) were benchmarked on small prototype datasets of 20 and 11 records using a 75:25 split. Ridge Regression gave the best range performance (MAE 7.7185 km, RMSE 10.7015 km, R^2 0.9927), and MultiOutput Gradient Boosting the best queue performance (queue-length MAE 0.6667 cars, waiting-time MAE 7.8805 minutes). The solution is a three-tier system comprising a React Native mobile application with seven screens, a FastAPI backend with 12 REST endpoints, a machine-learning inference microservice and MongoDB, with database and ML fallback mechanisms. Sixteen automated unit tests were reported as passed and ten system scenarios as verified. Because the hold-out sets contain only five and three records, the results are preliminary and demonstrate pipeline feasibility rather than real-world accuracy; no user study was conducted. The author's contribution comprises the model development and benchmarking, the ML microservice, the backend, the mobile application and the decision logic. Larger real-world datasets, live charger and traffic data, and user evaluation are required.

Keywords: Electric vehicles, range prediction, queue prediction, charging optimisation, machine learning

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
ARIMA	AutoRegressive Integrated Moving Average
CAN	Controller Area Network
CC-CV	Constant Current – Constant Voltage
EV	Electric Vehicle
JSON	JavaScript Object Notation
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
OBD-II	On-Board Diagnostics, second generation
OCPP	Open Charge Point Protocol
REST	Representational State Transfer
RMSE	Root Mean Squared Error
SaaS	Software as a Service
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SoC	State of Charge
UI	User Interface

1 INTRODUCTION

1.1 Overall project context

Electric vehicles (EVs) are widely regarded as an important part of the transition to more sustainable road transport. Their adoption, however, shifts several operational decisions onto the driver: how far the battery can actually take the vehicle, where and when to charge, how long a charging stop will take, and what it will cost. An over-optimistic range estimate, or a charging station with a long queue, can disrupt a journey and contribute to what is commonly described as range anxiety.

SmartEV is the group research project "Electric Vehicle Management System Using Artificial Intelligence and Machine Learning". It proposes an integrated, AI-powered platform that supports EV operation and decision-making. The wider project comprises several complementary components, including operational anomaly detection, battery health prediction with personalised recommendations, intelligent charging optimisation and range prediction, and a virtual assistant. This thesis addresses the individual component concerned with intelligent charging optimisation and range prediction.

1.2 Individual component context

The individual component converts vehicle, journey, station and user information into practical charging and travel guidance. As documented in the project report, it provides seven main functions: (i) driving-range prediction; (ii) charging-station queue and waiting-time forecasting; (iii) charging-station search and multi-criteria recommendation; (iv) route feasibility and charging-stop optimisation; (v) charging-duration estimation using a constant-current constant-voltage (CC-CV) model; (vi) charging-cost and monthly-savings analytics; and (vii) an integrated mobile dashboard.

The component is realised as a three-tier system: a React Native mobile application, a FastAPI backend, and a separate machine-learning (ML) inference microservice, supported by MongoDB with an in-memory fallback. Because predictions depend on the quality and coverage of their input data, the component presents its outputs as

estimates and is designed to degrade gracefully when the database or the ML service is unavailable.

1.3 Problem statement

EV drivers often lack a single decision-support tool that combines estimated remaining range, charging-station suitability, predicted waiting time, route options, charging duration and expected cost. When these functions are provided separately or rely on static information, a driver may select a station that is difficult to reach, congested, expensive or unsuitable for the vehicle, which creates uncertainty and reduces the practical value of EV applications.

The research problem is therefore how to design, implement and evaluate an integrated EV decision-support component that uses available historical, prototype and user-provided data to estimate range and charging queues and to support charging and route decisions.

1.4 Research gap

Existing EV applications commonly specialise in one function: routing, station discovery, or vehicle-specific range estimation. Range prediction, station information, queue estimation, route planning and cost analysis are therefore frequently treated as separate problems, whereas a driver's decision depends on all of them together. A recommendation must additionally combine predictions with constraints such as remaining range, connector compatibility, charging speed, station cost and user preferences.

The gap addressed here is the lack of an integrated, modular decision-support workflow that links ML-based range and queue prediction with station ranking, route feasibility, charging-time estimation and cost analytics, and that continues to operate when parts of the infrastructure are unavailable. The effectiveness of this integration is assessed through model benchmarking and system testing in Chapters 5 and 6; no user-based evaluation has been conducted.

1.5 Aim and objectives

1.5.1 Main aim

To develop and evaluate an AI-based intelligent EV charging optimisation and range prediction component that combines predictive analytics and decision logic to estimate driving range, forecast station queues, and support charging and route decisions through a mobile application.

1.5.2 Specific objectives

1. Develop and benchmark a range-prediction module using the available vehicle and driving features.
2. Develop and benchmark a queue-length and waiting-time prediction module using the available station-usage data.
3. Implement a charging-station search and multi-criteria recommendation mechanism.
4. Implement route feasibility checking and charging-stop optimisation.
5. Implement CC-CV charging-duration estimation and charging-cost and monthly-savings analytics.
6. Expose these capabilities through a FastAPI backend (12 REST endpoints) and an ML inference microservice, and present them in a seven-screen mobile application, with fallback behaviour when the database or ML service is unavailable.
7. Evaluate predictive performance, functional correctness, integration behaviour and the limitations of the prototype.

1.6 Research questions or hypotheses

The study is guided by the following research questions.

RQ1: Which model, among the benchmarked candidates, best predicts remaining EV driving range on the held-out data, measured by MAE, RMSE and R^2?

RQ2: Which model best predicts charging-station queue length and waiting time on the held-out data?

RQ3: How can range estimates, queue predictions, station attributes, route constraints and cost estimates be combined into charging recommendations within one workflow?

RQ4: To what extent does the implemented component satisfy its functional and non-functional requirements, including graceful behaviour under service failure?

Working hypothesis: data-driven models can learn useful relationships from suitable EV and station data, but their performance will vary with dataset size, feature availability and operating conditions. Because the prototype datasets are very small, this hypothesis can only be assessed preliminarily in Chapter 6.

1.7 Scope limitations and assumptions

Scope. The scope covers model development (range and queue), station search and ranking, route feasibility and charging-stop optimisation, charging-duration and cost estimation, backend and ML-service development, mobile-application integration, and technical evaluation.

Limitations. (i) The datasets are small prototype samples: 20 range records and 11 queue records. (ii) There is no direct CAN-bus or OBD-II vehicle-telemetry integration. (iii) There is no live Open Charge Point Protocol (OCPP) charger data. (iv) Dynamic traffic and elevation information is limited. (v) No formal driver-acceptance study was conducted. (vi) Screenshots of the mobile screens have not yet been captured in the documented workspace.

Assumptions. The prototype datasets are assumed to contain patterns relevant to the chosen prediction tasks, but this assumption has not been validated. The thesis draft reports 20 range records and 11 queue records; the provenance (public, simulated, collected or mixed), licence, collection period and consent basis were not available in the supplied file. These are PENDING and must be filled from the data files or project records before submission. Results are preliminary evidence of pipeline behaviour, not real-world accuracy.

1.8 Individual contribution

The individual contribution of the author (IT22134080) comprises: developing and benchmarking the range and queue prediction models; implementing the ML microservice and model-loading interfaces; developing the 12 backend API endpoints and their business logic; building the seven mobile application screens; implementing station ranking, route feasibility, charging-duration and cost analytics; and preparing the training notebooks, evaluation plots, structured evidence and dissertation drafts. The principal contribution is the integration of these capabilities into a modular system with fallback support. Work belonging to other group members, including the other components of SmartEV, is outside this thesis.

1.9 Thesis structure

Chapter 1 introduces the project, problem, gap, aims and scope. Chapter 2 reviews the technical foundations and related work. Chapter 3 describes the research design, data, procedure, metrics and ethical considerations. Chapter 4 presents the requirements and solution design. Chapter 5 documents implementation, integration and testing. Chapter 6 reports the results. Chapter 7 discusses the findings, contributions, limitations and implications, including commercial and sustainability considerations. Chapter 8 concludes the thesis and proposes future work. The references, glossary and appendices follow.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The review is a narrative overview organised around four research areas identified in the project report: (i) machine learning for EV range prediction; (ii) charging-demand and queue forecasting; (iii) non-linear EV charging and route optimisation; and (iv) multi-criteria charging-station selection. Foundational methods used in the implementation (regression, ensemble learning, graph search) are supported by their original publications..

2.2 Domain and theoretical foundations

2.2.1 EV range prediction

Remaining driving range is an estimate of the distance a vehicle can travel with its available energy under given operating conditions. Influencing factors commonly include battery state of charge (SoC), energy consumption, vehicle characteristics, speed, acceleration, road gradient, ambient temperature, traffic and auxiliary load [1]. The target (remaining range, in km in this work) and its units must be defined explicitly.

2.2.2 Regression and ensemble models

Regression estimates a continuous target. Ordinary linear regression fits coefficients by least squares; ridge regression adds an L2 penalty on the coefficients to stabilise estimates when the sample is small or the features are correlated [5]. Random forests average many decorrelated decision trees [6], and gradient boosting builds trees sequentially to correct residual errors [7]. All four models are implemented in scikit-learn [8]. The project proposal also names long short-term memory (LSTM) networks [9] as a candidate for sequence data; LSTM does not appear among the benchmarked models reported for this component.

2.2.3 Queue and waiting-time prediction

Waiting time at a charging station depends on arrival demand, number of chargers, charging durations and operating hours [2]. The proposal identifies ARIMA and LSTM as time-series candidates; their suitability depends on the quantity and regularity of timestamped data. In this work queue length and waiting time are

predicted jointly as a two-output regression problem using multi-output wrappers around Ridge, Random Forest and Gradient Boosting.

2.2.4 Station selection and route optimisation

A station recommendation should test feasibility (reachability, connector compatibility, availability) before preference (distance, speed, cost, queue) [4]. Graph search represents locations as nodes and travel connections as edges: Dijkstra's algorithm finds shortest paths for non-negative costs [10], and A* adds a heuristic to guide the search [11]. EV routing adds battery-feasibility constraints and charging-stop decisions [3]. If traffic and elevation data are absent, the route must not be described as traffic- or terrain-aware.

2.2.5 Charging-time and cost estimation

Lithium-ion packs are commonly charged with a CC-CV profile: current is held constant until the cell voltage reaches a limit, after which voltage is held constant and current tapers [12]. Charging power therefore falls at high SoC, and a CC-CV model gives a more realistic duration than dividing energy by nominal power. Cost is derived from the energy delivered and the applicable tariff.

2.3 Related research

The project report identifies the four research areas summarised in Table 2.1. It does not supply bibliographic details for individual studies, so this section records the comparison framework rather than study-level findings.

Table 2.1 Research areas, common approaches and comparison points

Research area	Common approaches	Comparison points for this thesis
Range prediction	Regression, ensemble learning, neural networks [1]	Target definition, input features, split strategy, MAE/RMSE/R^2
Queue and demand forecasting	Historical baselines, ARIMA, LSTM, regression [2]	Forecast horizon, granularity, MAE/RMSE
Charging and route optimisation	Dijkstra, A*, EV-aware non-linear routing [3]	Objective (distance/time), battery feasibility, charging stops
Station selection	Rule-based filtering, weighted multi-criteria scoring [4]	Compatibility, distance, availability, cost, explainability

2.4 Existing systems products and techniques

Official product documentation indicates different emphases. A Better Routeplanner (ABRP) plans EV trips, recommends charging stops, and describes use of vehicle-specific energy behaviour, elevation, weather and live charger availability [18]. PlugShare focuses on station discovery, user reviews, photos, compatibility filters and trip-planning features [19]. Tesla's Trip Planner can add Supercharger stops based on the vehicle's current state of charge and destination [20]. ChargePoint documents station discovery and a Route Planning flow with stops informed by EV range and state of charge; some advanced route-planning features are described as available only to select users [21]. These are feature-level comparisons based on official product pages, not performance tests. Availability and feature coverage in Sri Lanka were not verified, so regional availability is PENDING. SmartEV's intended integration of queue prediction, weighted station ranking, route feasibility and cost estimation is a prototype contribution, but the current evidence does not establish that it outperforms these products.

2.5 Gap synthesis and conceptual framework

Figure 2.1 shows the conceptual framework. Input data are validated and preprocessed, then feed the range and queue prediction modules. Station filtering and route-feasibility logic combine these predictions with station attributes and trip constraints. Charging-time and cost calculations add operational context, and the recommendation layer presents ranked options and alerts through the mobile interface.

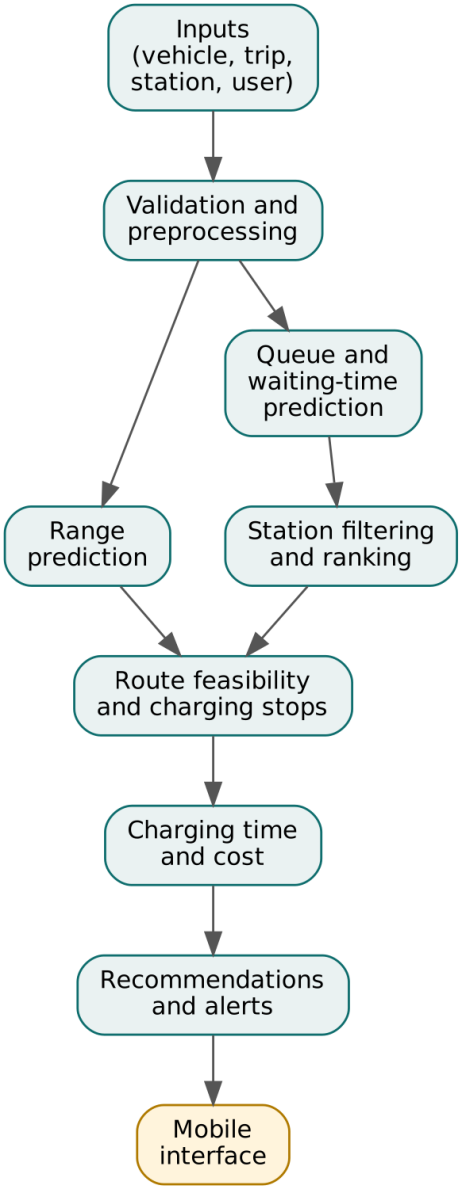


Figure 2.1 Conceptual framework of the charging optimisation and range prediction component

The contribution is assessed by whether these functions operate together coherently and whether the prediction modules show measurable performance on held-out data.

3 RESEARCH METHODOLOGY

3.1 Research design

The study follows a Design Science Research approach [13] combined with experimental evaluation. The artefact is an EV decision-support component. The process comprised problem identification, requirements definition, architecture design, data preparation, development of predictive and decision logic, integration, and evaluation. The evaluation addresses a research question (how well do the models predict range and queue behaviour on held-out data?) and an engineering question (does the integrated component provide the required functions reliably?).

3.2 Data participants or study materials

Two prototype datasets were used to train and validate the pipelines (Table 3.1). The report describes them as small samples intended to validate training and inference rather than to establish generalisability.

Table 3.1 Datasets used

Dataset	Records	Columns	Prediction target	Hold-out records (25%)
EV Range Dataset	20	8 columns (field names pending CSV verification)	Remaining driving range (km)	5

Dataset	Records	Columns	Prediction target	Hold-out records (25%)
EV Queue Dataset	11	9 columns; includes driving_mode (verify full schema against CSV)	Queue length (cars) and waiting time (min)	3

Because the hold-out sets contain only five (range) and three (queue) records, all metrics are preliminary and cannot be taken as evidence of real-world accuracy.

3.3 Experimental or study procedure

1. Load each dataset and split it 75:25 into training and test sets with random_state=42.
2. Train the candidate range models: Linear Regression, Ridge Regression, Random Forest and Gradient Boosting.
3. Train the candidate queue models as multi-output regressors: MultiOutput Ridge, MultiOutput Random Forest and MultiOutput Gradient Boosting, predicting queue length and waiting time jointly.
4. Evaluate every model on the same hold-out records using MAE, RMSE and (for range) R^2; for queue prediction report queue-length MAE, waiting-time MAE and overall RMSE.
5. Select a champion model per task by the stated criterion (lowest test MAE and RMSE for range; lowest overall RMSE and waiting-time MAE for queue).

6. Serialise the selected models as .joblib files with metadata and load them in the ML inference microservice.
7. Implement station search and ranking, route feasibility, CC-CV duration estimation and cost analytics as backend business logic.
8. Run 13 backend unit tests, 3 ML-microservice tests and 10 system-level evaluation scenarios, including fallback behaviour.
9. Record notebooks, evaluation plots and structured evidence for reproducibility.

3.4 Measures and evaluation metrics

For n test observations with observed values y and predictions ŷ, the following metrics are used.

- Mean Absolute Error: $MAE = (1/n) \Sigma |y - \hat{y}|$, expressed in the unit of the target (km, cars or minutes).
- Root Mean Squared Error: $RMSE = \sqrt{[(1/n) \Sigma (y - \hat{y})^2]}$, which penalises large errors more strongly [14].
- Coefficient of determination: $R^2 = 1 - SS_res / SS_tot$, reported for range prediction.

Lower MAE and RMSE indicate better predictions on the same target and test set. For the queue task, the supplied thesis reports an "overall RMSE" across two outputs, but the aggregation method and whether outputs were scaled are not documented; interpret that value cautiously until the evaluation code confirms its calculation. Precision and recall are not reported because no independent relevance labels exist for recommendations; recommendation logic is instead reported as verified through scenario-based tests.

3.5 Data analysis

Results are compared in model-comparison tables (Tables 6.1 and 6.2) and charts (Figures 6.1 and 6.2). Linear Regression serves as the simple baseline for range, and MultiOutput Ridge as the linear baseline for queue prediction. Relative differences are computed against the baseline. Because hold-out sets contain five and three records,

differences are interpreted descriptively; no significance tests or confidence intervals are reported, as they would not be meaningful at this sample size.

Station-filtering, recommendation, route, charging-duration and cost modules are verified by test cases and scenarios (Section 5.4) rather than by predictive metrics.

3.6 Ethics privacy security and responsible AI

No human participants were involved and no formal user study was conducted; therefore no institutional ethics approval or informed-consent procedure was required for the work reported. If future work collects driver or location data, user data should be anonymised, collection minimised, participation voluntary with informed consent, and access restricted, with secure transmission and storage. Secrets, API keys and personal data must be kept out of source control.

Responsible-AI considerations: predictions are presented as estimates, not guarantees; the system must not imply that it can guarantee a vehicle will reach a station; and outputs produced by fallback calculations should be distinguishable from model-based outputs. Use of generative AI in preparing this thesis is disclosed in Appendix B.

3.7 Reproducibility and research artefacts

Reproducibility is supported by training notebooks, evaluation plots and structured evidence produced during the project, a fixed random seed (random_state=42), a fixed 75:25 split, and serialised .joblib models with metadata.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Stakeholders

- EV owners and drivers who need range estimates and charging guidance.
- Charging-station operators whose station data may be displayed or consumed by the system.
- Fleet operators who may use charging and cost information to plan vehicle usage.
- Developers and administrators who maintain the application, models and data integrations.

4.1.2 Functional requirements

FR1: The system shall predict remaining driving range from vehicle and journey inputs and state the unit.

FR2: The system shall predict charging-station queue length and waiting time.

FR3: The system shall search and filter charging stations and rank them using multiple criteria.

FR4: The system shall assess route feasibility and recommend charging stops.

FR5: The system shall estimate charging duration using a CC-CV model.

FR6: The system shall compute charging cost and monthly savings analytics.

FR7: The system shall present all results through a mobile dashboard and dedicated feature screens.

FR8: The system shall validate inputs and return clear errors for invalid requests.

FR9: The system shall continue to operate through fallback mechanisms when the database or ML service is unavailable.

4.1.3 Non-functional requirements

Table 4.1 lists the non-functional requirements, their design implications and the verification approach.

Table 4.1 Non-functional requirements

Requirement	Design implication	Verification approach
Usability	Clear labels, units and understandable recommendations; seven task-focused screens	Developer testing; formal user study not yet done
Performance	Asynchronous ML calls; separate inference service	Response-time measurement
Reliability	Input validation; database and ML fallbacks	Negative and failure-path tests
Maintainability	Modular three-tier architecture; documented REST interface	Module-level unit tests
Scalability	ML inference separated from the application backend	Concurrent-request testing still required
Security and privacy	Protected secrets; minimised personal data	Configuration review

4.2 Component architecture within the complete solution

The component follows a modular, three-tier microservice architecture (Figure 4.1). The React Native application (presentation tier) calls the FastAPI backend over HTTP/JSON. The backend (application tier) validates requests, applies business logic (ranking, route feasibility, CC-CV duration, cost analytics) and asynchronously calls the ML inference microservice (intelligence tier), which runs on port 8001 and loads serialised .joblib models with metadata. Persistent data are held in MongoDB (data tier), with an in-memory fallback if the database is unavailable. If the ML service is unavailable, the backend uses fallback calculations.

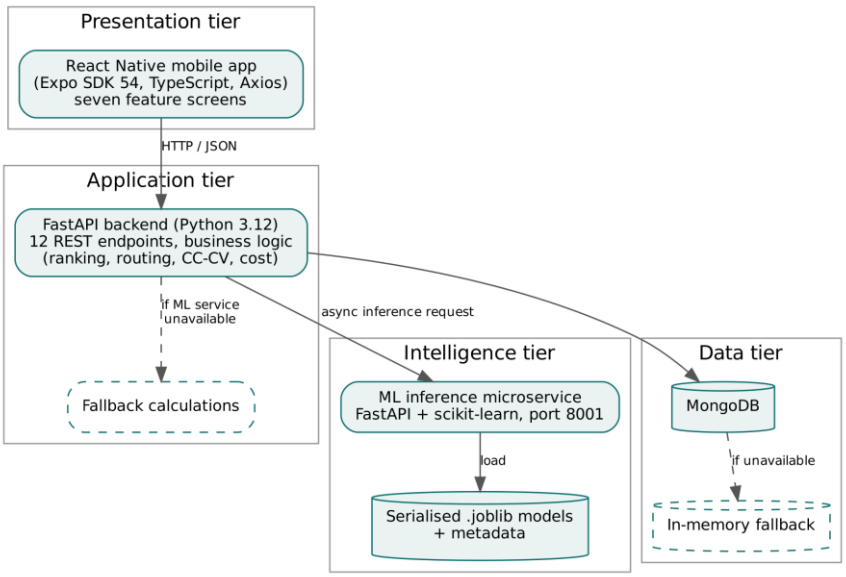


Figure 4.1 System architecture of the component

Within the wider SmartEV platform, this component supplies range, queue, station, route and cost services to the shared mobile application. Interfaces with the other group components are discussed in Section 5.3.

4.3 Design alternatives and decisions

Table 4.2 summarises the main design decisions and the evidence for each.

Table 4.2 Design alternatives and decisions

Decision area	Candidates	Decision and rationale
Range model	Linear, Ridge, Random Forest, Gradient Boosting (LSTM proposed)	Ridge selected: lowest test MAE (7.7185 km) and RMSE (10.7015 km) of the four benchmarked models
Queue model	MultiOutput Ridge, Random Forest, Gradient Boosting (ARIMA/LSTM proposed)	MultiOutput Gradient Boosting selected: lowest overall RMSE (5.1456) and waiting-time MAE (7.8805 min)
Station ranking	Rule-based filtering and multi-criteria scoring	Multi-criteria recommendation implemented
Service deployment	Integrated backend or separate prediction service	Separate FastAPI/scikit-learn ML microservice for modularity and independent model loading
Database	MongoDB with or without fallback	MongoDB with in-memory fallback for resilience
Mobile framework	React Native or other cross-platform tools	React Native with Expo SDK 54 and TypeScript

The proposal considered Node.js for the backend and LSTM/ARIMA for forecasting; the reported implementation instead uses Python FastAPI and scikit-learn models. This is consistent with the documented prototype and its small datasets (20 range records and 11 queue records), although the supplied evidence does not include a recorded decision log. Deep sequence models and time-series methods are therefore treated as proposed alternatives, not implemented or benchmarked methods.

4.4 Detailed design

4.4.1 Range-prediction flow

The mobile app submits vehicle and journey inputs; the backend validates them and forwards a request to the ML microservice, which applies the preprocessing used at training and runs the Ridge model to return the predicted remaining range in kilometres. If the ML service is unreachable, the backend returns a fallback calculation, which should be identified as such in the response.

4.4.2 Queue-prediction flow

Station and time features are sent to the ML microservice, where the MultiOutput Gradient Boosting model returns the predicted queue length (cars) and waiting time (minutes) for the station.

4.4.3 Station search, ranking and route feasibility

The backend first filters stations using the hard criteria supported by the implementation and then applies a weighted score. The corrected weights specified for this dissertation are distance 0.25, availability 0.20, charging power 0.20, cost 0.15, rating 0.10 and predicted queue/wait 0.10; the weights sum to 1.00. The score should be interpreted as a decision-support ranking, not a validated probability of station quality. The exact distance provider/route graph implementation could not be independently inspected from the files available for this revision and remains pending source-code confirmation.

4.4.4 Charging duration and cost

Charging duration is described as a CC-CV estimate, with a constant-current phase followed by a constant-voltage taper phase. Cost analytics are described as multiplying energy delivered by the applicable tariff. The exact transition SoC, charging efficiency, tariff source/rate and any monthly-savings assumptions were not present in the supplied thesis evidence and are not asserted here. No tariff rate or measured charging duration is invented.

For intuition only, consider an illustrative (not system-generated) case: a 40 kWh battery charged from 20% to 80% requires $0.6 \times 40 = 24$ kWh. At a constant effective power of 50 kW × 0.9 = 45 kW, the simple estimate is $24 \div 45 \approx 0.53$ h (about 32 min). A CC-CV model gives a longer duration because power tapers at high SoC, which is why the system does not rely on the constant-power estimate.

4.4.5 API and interface design

The backend exposes 12 REST endpoints grouped as shown in Table 4.3, and the mobile application provides seven screens: Dashboard, Range Prediction, Charging Stations, Queue Prediction, Route Optimization, Charging Estimator and Cost Analysis.

Table 4.3 Backend API functional groups

Function group	Purpose	Method and path
Range prediction	Predict remaining driving range	[required]
Queue prediction	Predict queue length and waiting time	[required]
Station search and recommendation	Search, filter and rank stations	[required]
Route optimisation	Check feasibility and propose charging stops	[required]
Charging estimate	Estimate CC-CV charging duration	[required]
Cost analytics	Charging cost and monthly savings	[required]

4.5 Standards and professional practice

The implementation uses modular code, input validation, error handling with fallback behaviour, serialised models with metadata, and documented API specifications.

Dataset licences and third-party API terms must be respected, secrets and personal information protected, and the use of generative AI disclosed under SLIIT policy (Appendix B).

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Table 5.1 lists the technology stack reported for the implementation.

Table 5.1 Technology stack

Layer	Technology
Mobile frontend	React Native, Expo SDK 54, TypeScript, Axios
Backend	Python 3.12, FastAPI
ML microservice	FastAPI and scikit-learn (port 8001)
Database	MongoDB with in-memory fallback
Model storage	Serialised .joblib models with metadata
Documentation	Research notebooks, experimental evidence, API specifications, dissertation drafts

The stack reported in the thesis includes React Native, Expo SDK 54, TypeScript, Axios, Python 3.12, FastAPI, scikit-learn, joblib and MongoDB. Exact installed versions for every library, the test framework, lockfile and runtime environment were not included in the supplied file and remain PENDING; do not infer them from current documentation.

5.2 Component implementation

5.2.1 Data preparation and model training

Both datasets are split 75:25 with random_state=42 (15 training and 5 test records for range; 8 training and 3 test records for queue). Candidate models are trained, evaluated on the same hold-out records, and the champions are serialised as .joblib files with metadata for loading by the ML microservice.

5.2.2 Range prediction

Four models were benchmarked and Ridge Regression was selected as the range champion. The ML microservice loads the serialised model and returns a prediction in kilometres through the backend. Fallback calculations are used if the service is unavailable.

5.2.3 Queue prediction

A MultiOutput Gradient Boosting model predicts queue length and waiting time jointly. The proposal's ARIMA and LSTM options are not among the reported benchmarks; the small queue dataset (11 records) offers no temporal structure sufficient to evaluate such models, which should be stated as a limitation.

5.2.4 Station ranking, routing, charging duration and cost

The backend implements station search with multi-criteria recommendation, route feasibility and charging-stop optimisation, CC-CV charging-duration estimation, and cost and monthly-savings analytics. These modules include input validation and fallback behaviour.

5.2.5 Mobile application and API integration

The Expo/React Native app uses Axios to call the backend and provides seven feature screens (Dashboard, Range Prediction, Charging Stations, Queue Prediction, Route Optimization, Charging Estimator, Cost Analysis).

5.3 Integration with the other components

Within this component, integration covers the mobile application, FastAPI backend, ML inference microservice and MongoDB, including asynchronous ML calls, model-loading mechanisms and the database and ML fallbacks. The supplied materials do not document integration with the other SmartEV components (anomaly detection, battery health, virtual assistant).

5.4 Verification and software testing

Verification combined automated unit tests with system-level scenarios. Backend unit tests (13) and ML-microservice tests (3) give 16 automated tests, all reported as passed. Ten system evaluation scenarios were reported as verified. Coverage areas are summarised in Table 5.2.

Table 5.2 Test coverage areas and reported outcomes

Area	Test level	Reported outcome
Range prediction	Unit / system	Covered; no failures reported
Queue forecasting	Unit / system	Covered; no failures reported

Area	Test level	Reported outcome
Station filtering	Unit / system	Covered; no failures reported
Recommendation ranking	Unit / system	Covered; no failures reported
Route feasibility	Unit / system	Covered; no failures reported
Charging duration	Unit / system	Covered; no failures reported
Cost calculation	Unit / system	Covered; no failures reported
Input validation	Unit / system	Covered; no failures reported
Fallback behaviour	System	Covered; no failures reported

5.5 Deployment and operation

The prototype consists of the Expo mobile app, the FastAPI backend, the ML microservice on port 8001 and a MongoDB instance (with in-memory fallback). It is a development prototype; no production deployment is claimed.

6 RESULTS AND VALIDATION

6.1 Research results

6.1.1 Range prediction performance

Table 6.1 and Figure 6.1 compare the four range models on the same five hold-out records (target: remaining range in km).

Table 6.1 Range-prediction results on the hold-out set (n = 5)

Model	MAE (km)	RMSE (km)	R^2
Linear Regression (baseline)	8.0697	10.8125	0.9925
Ridge Regression (selected)	7.7185	10.7015	0.9927
Random Forest	37.9834	49.2332	0.8446
Gradient Boosting	50.0683	57.4954	0.7881

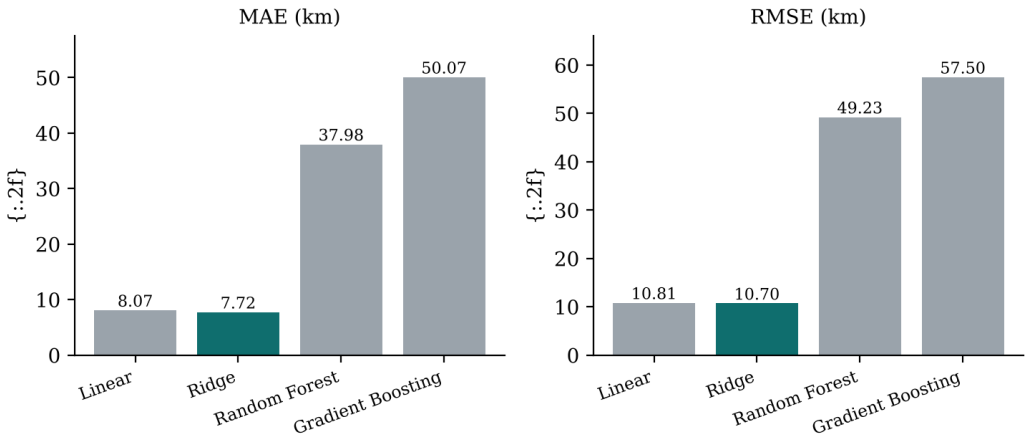


Figure 6.1 Range-prediction MAE and RMSE by model (selected model highlighted)

Ridge Regression achieved the lowest MAE and RMSE and was selected as the range champion. Its MAE is 0.35 km (4.4%) lower than the Linear Regression baseline and its RMSE 0.11 km lower; this margin is small and, given five test records, should not be treated as a reliable difference. The tree ensembles were markedly worse: Random Forest's MAE was about 4.9 times, and Gradient Boosting's about 6.5 times, that of Ridge.

6.1.2 Queue prediction performance

Table 6.2 Queue-prediction results on the hold-out set (n = 3)

Model	Queue-length MAE (cars)	Waiting-time MAE (min)	Overall RMSE
MultiOutput Ridge (baseline)	0.8117	11.4652	7.4358
MultiOutput Random Forest	0.8167	14.3333	7.8424
MultiOutput Gradient Boosting (selected)	0.6667	7.8805	5.1456

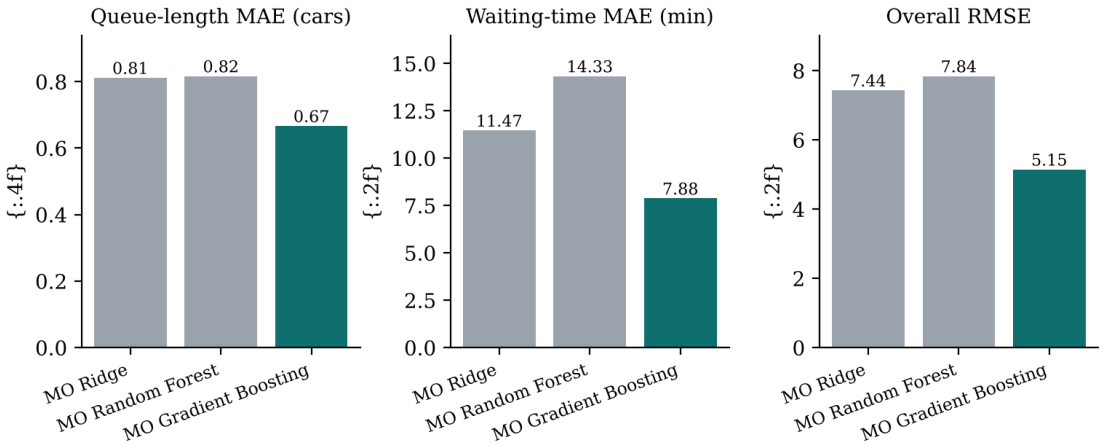


Figure 6.2 Queue-prediction errors by model (selected model highlighted)

MultiOutput Gradient Boosting gave the lowest errors on all three reported measures (Table 6.2, Figure 6.2). Relative to the Ridge baseline, its waiting-time MAE was 3.58 minutes (31.3%) lower and its queue-length MAE 0.15 cars (17.9%) lower. With only three test records, these results are preliminary.

6.1.3 Recommendation and route scenarios

Recommendation ranking, station filtering and route feasibility were verified through the system evaluation scenarios (10 of 10 reported as verified) rather than through precision or recall.

6.2 Product and technical results

Table 6.3 summarises the implemented features. Quantities (12 endpoints, 7 screens) are as reported.

Table 6.3 Feature status

Feature	Status	Evidence	Known limitation
Range prediction	Implemented (Ridge)	Benchmark; unit tests	20-record dataset
Queue prediction	Implemented (MO Gradient Boosting)	Benchmark; unit tests	11-record dataset; no live OCPP data
Station search and recommendation	Implemented	Tests; scenarios	No live availability data
Route feasibility and charging stops	Implemented	Tests; scenarios	Limited traffic and elevation data
CC-CV charging estimate	Implemented	Tests; scenarios	Not validated against real chargers
Cost and savings analytics	Implemented	Tests; scenarios	Tariff data assumptions
Mobile dashboard (7 screens)	Implemented	Application build	Screenshots not yet captured
Fallback mechanisms	Implemented	Fallback scenarios verified	—

Runtime clarification: the reported 0.96 ms and 49.84 ms figures refer to model training durations only. The reported 9.198 seconds is the total test-suite execution time. It must not be described as API latency or inference latency. No measured API response latency was available in the supplied evidence.

Response times were not reported and are not claimed.

6.3 User or stakeholder validation

No formal driver-acceptance or user study was conducted. Usability has therefore been assessed only through developer testing, and no participant data, survey results or satisfaction measures are reported. A user study, with appropriate consent and institutional approval, is identified as future work.

6.4 Integration results

Integration was verified at three levels: 13 of 13 backend unit tests, 3 of 3 ML-microservice tests and 10 of 10 system scenarios (Table 6.4). The scenarios included

fallback behaviour when the database or ML service is unavailable. Concurrent-request behaviour and live external data integration were not tested.

Table 6.4 Reported test outcomes

Test set	Passed / total
Backend unit tests	13 / 13
ML microservice tests	3 / 3
System evaluation scenarios	10 / 10

6.5 Summary of findings

The reported 48-item requirements audit classified 43 items (89.6%) as VERIFIED, 4 (8.3%) as PARTIALLY VERIFIED and 1 (2.1%) as NOT AVAILABLE. These figures do not mean that 48/48 items are complete. The identities of the four partial items and one unavailable item were not supplied in the thesis evidence available for this revision and must be mapped to the audit checklist. Screenshots, final front matter and evidence traceability remain outstanding.

Table 6.5 Requirements audit

Status	Items	Percentage
Verified	43	89.6%
Partially verified	4	8.3%
Not available	1	2.1%

Regarding the research questions: RQ1 and RQ2 are answered preliminarily (Ridge for range; MultiOutput Gradient Boosting for queue) on very small hold-out sets; RQ3 is answered by the implemented integrated workflow; RQ4 is supported by the passed tests and audit but not by concurrency, real-data or user testing. The evidence does not establish real-world accuracy or generalisation.

7 DISCUSSION

7.1 Interpretation of findings

The strong R^2 (0.9927) and low error of Ridge Regression on the range task suggest a largely linear relationship in the small dataset, which a regularised linear model captures well. A plausible, but untested, explanation for the poor tree-ensemble results is that with 15 training records tree-based models cannot extrapolate beyond observed values, whereas linear models can. Similarly, Gradient Boosting's advantage on the queue task may reflect its ability to model non-linear interactions in a small sample, but with three test records this is not conclusive. The close performance of Linear and Ridge Regression (R^2 0.9925 vs 0.9927) indicates that regularisation brought little benefit here.

These observations are directly supported by the experiments; the causal explanations require further investigation with larger datasets, cross-validation and feature-importance analysis.

7.2 Comparison with literature baselines and existing solutions

No directly comparable numerical baseline is reported for the SmartEV datasets. Relevant published work provides methodological context, but its metrics should not be compared directly with this prototype because the datasets, targets and evaluation protocols differ. The verified literature and official product documentation added to this revision are used for qualitative comparison only. SmartEV's potential distinction is the integration of range prediction, queue forecasting, station ranking, route feasibility, charging-duration estimation and cost analytics; this is a feature-level comparison, not a claim of superiority.

7.3 Research contribution

The research contribution is an evaluated workflow connecting range estimation and queue forecasting with station ranking, route feasibility, charging-duration estimation and cost analytics, together with a benchmark of seven candidate models on prototype data. The strength of the contribution is limited by dataset size; its main value at this

stage is a validated, reproducible pipeline that can be retrained on larger real-world data.

7.4 Product contribution

The product contribution is a working prototype: a FastAPI backend with 12 REST endpoints, an ML microservice, and a seven-screen React Native application, with database and ML fallback mechanisms and 16 automated tests plus 10 verified system scenarios. Features not completed include live OCPP/traffic data and CAN-bus or OBD-II telemetry.

7.5 Limitations and threats to validity

- Dataset size: 20 range and 11 queue records, with hold-out sets of 5 and 3, make metrics highly variable and unsuitable for generalisation.
- Dataset provenance: representativeness across vehicle types, regions and charging behaviour is unknown; provenance is not stated in the supplied report.
- Data limitations: no CAN-bus/OBD-II telemetry, no live OCPP data and limited traffic and elevation information.
- Model validation: a single random 75:25 split was used, without cross-validation or confidence intervals.
- Evaluation limitations: no concurrency testing and no user acceptance study.
- Temporal validity: charging demand, tariffs and traffic change over time.
- Documentation: UI screenshots and complete test traces are not yet available.

7.6 Implications

Implications for users, developers and researchers. Clear estimates with stated uncertainty could support more informed charging choices. For developers, the work shows the value of modular services and fallbacks. For researchers, validation on larger, representative datasets is the key next step.

Commercial and sustainability considerations. The primary users are individual EV owners; fleet operators, charging-station operators and mobility-service providers are possible secondary users. The value proposition is to combine range awareness, station suitability, waiting time, routing, duration and cost in a single workflow. The proposal

suggests a Software-as-a-Service model (individual subscriptions, fleet licensing, partnerships, premium analytics); these are potential models, not demonstrated revenue, and market claims would require verified market analysis and cost estimates for cloud, map APIs, data and compliance. A modular design allows services to be maintained separately, but scalability must be demonstrated by load tests. Code ownership, open-source licences, third-party API terms and the protection of location and vehicle data must be managed.

The project relates to UN Sustainable Development Goal 7 (Affordable and Clean Energy) and Goal 11 (Sustainable Cities and Communities) [16], through potentially more efficient use of charging infrastructure and better journey planning. These are potential contributions; no environmental benefit has been quantified. (The template's chapter list contains no separate commercialisation chapter, so this material is included here.)

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This thesis designed, implemented and evaluated an intelligent EV charging optimisation and range prediction component for the SmartEV platform. The prototype integrates ML-based range prediction, queue and waiting-time forecasting, station recommendation, route feasibility, CC-CV charging-duration estimation and cost analytics in a three-tier architecture with database and ML fallbacks.

On very small prototype datasets, Ridge Regression performed best for range (MAE 7.7185 km, RMSE 10.7015 km, R^2 0.9927) and MultiOutput Gradient Boosting for queue prediction (queue-length MAE 0.6667 cars, waiting-time MAE 7.8805 min). The system provides 12 REST endpoints and seven mobile screens and passed 16 automated tests and 10 system scenarios. These results are preliminary and demonstrate pipeline feasibility rather than real-world accuracy.

8.2 Achievement of objectives

Table 8.1 Achievement of objectives

Objective	Evidence	Achievement
1 Range prediction	Four models benchmarked; Ridge selected	Achieved (preliminary data)
2 Queue prediction	Three models benchmarked; MO Gradient Boosting selected	Achieved (preliminary data)
3 Station recommendation	Implemented; tests and scenarios	Achieved
4 Route feasibility and charging stops	Implemented; tests and scenarios	Achieved
5 CC-CV duration and cost analytics	Implemented; tests and scenarios	Achieved
6 Backend, ML service and mobile app with fallback	12 endpoints, 7 screens, fallbacks verified	Achieved
7 Evaluation	16 tests, 10 scenarios, 48-item audit (89.6% verified)	Partially achieved: no concurrency, real-data or user evaluation

8.3 Individual research and product contributions

The author developed and benchmarked the range and queue models; implemented the ML microservice and model loading; built the 12 backend endpoints and business logic (station ranking, route feasibility, charging duration, cost analytics); developed the seven mobile screens; and prepared notebooks, evaluation plots, structured evidence and drafts. The artefacts are the trained .joblib models, the backend and ML service, the mobile application and the evaluation evidence.

8.4 Future directions

- Collect thousands of real-world EV telemetry records and retrain and validate the models, using cross-validation.
- Integrate live charging-station availability data (e.g. OCPP).
- Incorporate real-time traffic and road elevation into routing.
- Conduct user acceptance studies with appropriate consent and institutional approval.
- Introduce Explainable AI techniques such as SHAP [17].
- Perform concurrency and stress testing and broader model validation.
- Complete the thesis evidence: screenshots, verified references and supervisor feedback.

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Evidence status convention: VERIFIED means supported by the thesis evidence or a checked public source; PENDING means the underlying repository, data, log or approval was not available for this revision; SCHEDULED means future work, not completed work.

Status summary (revision audit): VERIFIED — reported benchmark values, the 20/11-record dataset sizes, 75:25 split, random_state=42, 13 backend tests, 3 ML-service tests, 10 system scenarios, 12 API endpoint count and seven screen names as stated in the supplied thesis. PENDING — raw source-code inspection, CSV schema/provenance, endpoint-level contracts, full test logs, exact dependencies, audit item mapping, screenshots, CC-CV parameter evidence and supervisor sign-off. SCHEDULED — larger real-world datasets, cross-validation, concurrency/stress tests, live OCPP/traffic integration and formal user evaluation.

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GLOSSARY

Charging station – A facility where an EV can recharge its battery.

CC-CV charging – A charging profile with a constant-current phase followed by a constant-voltage phase in which current tapers.

Queue waiting time – Estimated time a vehicle waits before a compatible charger becomes available.

Range prediction – An estimate of the remaining distance an EV can travel under stated conditions.

State of Charge (SoC) – The battery's current charge level, usually as a percentage.

Baseline – A simple method used as a reference for evaluating other models.

Hold-out set – Records withheld from training and used only to evaluate a model.

Fallback mechanism – An alternative calculation or store used when the primary service is unavailable.

MultiOutput regression – A regression approach predicting more than one target at once.

Range anxiety – A driver's concern that the EV will run out of charge before reaching a charger or destination.

APPENDICES

Appendix A Research instruments or extended results

This appendix records the extended results and the information still required to complete the experiment record.

Table A.1 Full benchmark values reported

Task	Model	Metrics
Range	Linear Regression	MAE 8.0697; RMSE 10.8125; R^2 0.9925
Range	Ridge Regression	MAE 7.7185; RMSE 10.7015; R^2 0.9927
Range	Random Forest	MAE 37.9834; RMSE 49.2332; R^2 0.8446
Range	Gradient Boosting	MAE 50.0683; RMSE 57.4954; R^2 0.7881
Queue	MultiOutput Ridge	Queue MAE 0.8117; wait MAE 11.4652; RMSE 7.4358
Queue	MultiOutput Random Forest	Queue MAE 0.8167; wait MAE 14.3333; RMSE 7.8424
Queue	MultiOutput Gradient Boosting	Queue MAE 0.6667; wait MAE 7.8805; RMSE 5.1456

Experiment configuration reported in the thesis: 75:25 split; random_state=42; hold-out sizes 5 (range) and 3 (queue). Dataset provenance/licence, complete data dictionary, cleaning log, hyperparameters, exact package versions, all 12 endpoint contracts and per-test traces remain PENDING because they are not present in the supplied file. Confirm the data fields against the actual CSV files before final submission; the queue dataset feature is named driving_mode (not a similarly named substitute).

Appendix B Academic integrity and AI use disclosure

Tools used. The prior draft records use of Claude (Anthropic) to help assemble thesis text, structure chapters, draft prose, generate figures from reported values and format the document. ChatGPT was used in this revision to help edit and audit the dissertation, identify inconsistencies, draft text and organise a verification checklist. Web search was used to verify selected publication metadata and official product/software documentation. These statements

describe assistance in document preparation; the student must amend them if any tool or use is inaccurate and comply with SLIIT/module policy.

Verification performed. Existing benchmark values and test totals were retained from the supplied thesis draft unless explicitly corrected in this revision. The reported 0.96 ms and 49.84 ms values are model training durations, not inference latency. The value 9.198 seconds is the total test-suite execution duration only; it is not API latency. No API response-time measurements were available. Selected literature and official product pages were checked for bibliographic/product claims, but the source repository, raw datasets, test logs and project analysis report were not available in the uploaded files for line-by-line verification. The student must validate every claim against the repository and original sources.

Sections affected. All chapters, front matter and figures.

Outstanding evidence: genuine screenshots of the seven mobile screens; full test traces and the test-case inventory; dataset provenance/licences and field-level data dictionaries; exact endpoint contracts; exact dependency versions; identification of the audit's four partially verified and one unavailable items; source commit/hash; supervisor feedback; and completed front matter. No screenshots are fabricated or embedded by this revision.